

SC1001: Introduction to Computational Thinking

Week 5: Strings and Data Structures

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Slides

Slides and updates will be posted at

`boydgraber.org/teaching/SC_1001/`

Overview

1. Predict the output of common string operations
2. Trace how list methods change a list
3. Read slices and indexing from a list of lists
4. Build list comprehensions step by step
5. Use a dictionary as a small book database
6. Find palindromes in a sentence
7. Translate text into B-Sprache

Discussion 1: What Is the Output?

```
sample_string = "Hello, World! Welcome to Python programming."
upper_string = sample_string.upper()
print(f"Uppercase: {upper_string}")
lower_string = sample_string.lower()
print(f"Lowercase: {lower_string}")
capitalized_string = sample_string.capitalize()
print(f"Capitalized: {capitalized_string}")
position = sample_string.find("World")
print(f"Position of 'World': {position}")
replaced_string = sample_string.replace("World", "Universe")
print(f"Replaced string: {replaced_string}")
words_list = sample_string.split()
print(f"List of words: {words_list}")
joined_string = " ".join(words_list)
print(f"Joined string: {joined_string}")
```

Discussion 2: What Is the Output?

```
temps = [72, 68, 75, 70, 69, 74, 73]
print("start", temps)
temps.sort()
print("sort ", temps)
new_temp = 71
temps.append(new_temp)
print("add  ", temps)
print("max  ", max(temps))
print("min  ", min(temps))
pre = [51, 53, 56]
pre.append(temps)
print("pre  ", pre)
aft = [95, 97, 94]
for temp in aft:
    temps.append(temp)
print("week2", temps)
print("avg  ", sum(temps) / len(temps))
```

Output

Predict before running.

Discussion 3: What Is the Output?

- `pre.append(temps)` producted `[51, 53, 56, [68, 69, 70, 72, 73, 74, 75, 71]]`
- What if we wanted `[51, 53, 56, 68, 69, 70, 72, 73, 74, 75, 71]`?

Discussion 3: What Is the Output?

- `pre.append(temps)` produced `[51, 53, 56, [68, 69, 70, 72, 73, 74, 75, 71]]`
- What if we wanted `[51, 53, 56, 68, 69, 70, 72, 73, 74, 75, 71]`?

```
pre + temps
```

Discussion 3: What Is the Output?

```
weekly_schedule = [  
    ["Meeting", "Emails", "Project Work", "Lunch",  
     "Code Review", "Call with Client", "Report Writing"],  
    ["Code", "Emails", "Design", "Lunch",  
     "Testing", "Meeting", "Documentation"],  
    ["Research", "Emails", "Development", "Lunch",  
     "Planning", "Team Meeting", "Presentation"],  
]  
  
l1 = weekly_schedule[0][:3]  
print(l1)  
l2 = weekly_schedule[1][-3:]  
print(l2)  
l3 = weekly_schedule[2][4]  
print(l3)
```

Output

Sublists or one item?

Discussion 4: Squares of Numbers Divisible by Three

Goal: in as few lines as possible, create a list of all square numbers x^2 with $1 \leq x \leq 100$ that are divisible by 3.

- All numbers between 1 and 100
- Multiples of three
- Squares of all numbers between 1 and 100
- Put it together

Discussion 5: Book Database Setup

```
library = {  
  "978-1786074126": {"title": "How We Disappeared",  
                    "author": "Jing-Jing Lee",  
                    "year": 2019, "genre": "Historical Fiction"},  
  "978-1509855315": {"title": "Ponti",  
                    "author": "Sharlene Teo",  
                    "year": 2018, "genre": "Fiction"},  
  "978-0063161603": {"title": "Now You See Us",  
                    "author": "Balli Kaur Jaswal",  
                    "year": 2020, "genre": "Mystery"}  
}
```

Aside on ISBN numbers

- First three digits makes it a valid bar code: 978
- Next digits encode language / location
- Next digits encode publisher
- Whatever is left (except for last digit) identifies book
- Last digit is checksum

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 - ▶ 978-**0/1**: English language
 - ▶ 978-**2**: French language
 - ▶ 978-**3**: German language
 - ▶ 978-**4**: Japan (nation)
 - ▶ 978-**5**: CIS (group of nations)
 - ▶ 978-**65**: Brazil (nation)
 - ▶ 978-**7**: PRC (nation)
 - ▶ 978-**981**: Singapore (nation)
- Next digits encode publisher
- Whatever is left (except for last digit) identifies book
- Last digit is checksum

Aside on ISBN numbers

- First three digits makes it a valid bar code: 978
- Next digits encode language / location
- Next digits encode publisher
 - ▶ 978-1**01**: Penguin
 - ▶ 978-1**4012**: DC Comics
- Whatever is left (except for last digit) identifies book
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```
total = 0
for idx, digit in enumerate(int(x) for x in "945050147621"):
    # Multiply by 1 for even indices (0, 2, ...), 3 for odd indices (1, 3, ...)
    total += digit * (1 + 2 * (idx % 2))
    print(total, digit, 1 + 2 * (idx % 2))

check_digit = (10 - total % 10) % 10 # Mod 10 again in case 0
print(check_digit)
```

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```

- Similar to how Unicode characters are determined

Discussion 5A: Print the Whole Database

Write code to print each ISBN together with its title.

Discussion 5B: Look Up One ISBN

Look up ISBN 978-9815105866 and print the book information.

Discussion 5C: Add a New Book

Add *Raffles Hotel* by Ryu Murakami, published in 1989, in the genre Mystery.

Discussion 5D: Remove a Book

Remove ISBN 978-1509842794 from the database and print the removed entry.

Discussion 5E: Update a Field

Update the title for *Raffuruzu Hoteru* to Raffles Hotel, then print the updated entry.

Discussion 5F: Search by Author

Find all books by Jing-Jing Lee.

Discussion 5G: Filter by Genre

Create a new dictionary containing only the books in the genre `Mystery`.

Discussion 6: Palindromes in a Sentence

Given a sentence, output all words that are palindromes.

```
sentence = "level civic python radar banana madam code"
```

Discussion 7: B-Sprache

Translate each word into B-Sprache by doubling every vowel with a b in between.

Example: arbeit → abarbebeibit

Takeaways

- String methods transform text.
- List methods can change a list in place or create nested structure.
- Slices return sublists, while indexing returns one element.
- List comprehensions combine generation, transformation, and filtering.
- Dictionaries work well for structured records (like json).
- String processing problems often combine loops, conditions, and membership tests.